FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4

MATHEMATICS

MAT5CJ301—REAL ANALYSIS II

Module III: Properties of the Riemann Integral, Cauchy Criterion, Integrability & Fundamental Theorems

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Suppose that f and g are in $\mathcal{R}[a,b]$. If $f(x) \leq g(x)$ for all $x \in [a,b]$, prove that $\int_a^b f \leq \int_a^b g$.	**3**
2	If $f \in \mathcal{R}[a,b]$ and $\lvert f(x)\rvert \leq M$ for all $x \in [a,b]$, show that $\left\lvert \int_a^b f \right\rvert \leq M(b-a)$.	**3**
3	State Cauchy's Criterion and the Squeeze Theorem for the Riemann integrability of a function $f : [a,b] \to \mathbb{R}$.	**3**
4	Define a step function on $[a,b]$. Let $g : [0,4] \to \mathbb{R}$ be defined by $g(x) = 3$ for $0 \leq x \leq 2$ and $g(x) = 2$ for $2 < x \leq 4$. Evaluate $\int_0^4 g$.	**3**
5	Suppose that $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, $f(x) \geq 0$ for all $x \in [a,b]$, and $\int_a^b f = 0$. Prove that $f(x) = 0$ for all $x \in [a,b]$.	**3**
6	If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ with $a < b$, prove that there exists $c \in [a,b]$ such that $\int_a^b f = f(c)(b-a)$.	**3**
7	State the First Form of the Fundamental Theorem of Calculus. Explain why it cannot be applied to $H(x) = 2\sqrt{x}$ and $h(x) = H'(x) = \frac{1}{\sqrt{x}}$ on $[0,b]$.	**3**
8	Using the First Form of the Fundamental Theorem of Calculus, show that there does not exist a continuously differentiable function f on $[0,2]$ such that $f(0) = -1$, $f(2) = 4$, and $f'(x) \leq 2$ for $0 \leq x \leq 2$.	**3**
9	Define the indefinite integral F of $f \in \mathcal{R}[a,b]$ with basepoint a. If $\lvert f(x)\rvert \leq M$ for all $x \in [a,b]$, prove that $\lvert F(z) - F(w)\rvert \leq M\lvert z-w\rvert$ for all $z, w \in [a,b]$.	**3**
10	State Lebesgue's Integrability Criterion. Using the Composition Theorem, prove that if f and g belong to $\mathcal{R}[a,b]$, then the product fg belongs to $\mathcal{R}[a,b]$.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	State and prove the Boundedness Theorem for Riemann integrable functions: If $f \in \mathcal{R}[a,b]$, then f is bounded on $[a,b]$.	**6**
12	Suppose that f and g are in $\mathcal{R}[a,b]$. Prove that: (a) if $k \in \mathbb{R}$, the function kf is in $\mathcal{R}[a,b]$ and $\int_a^b kf = k\int_a^b f$; (b) the function $f+g$ is in $\mathcal{R}[a,b]$ and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.	**6**
13	Define an elementary step function on $[a,b]$. Prove that if $\varphi : [a,b] \to \mathbb{R}$ is any step function, then $\varphi \in \mathcal{R}[a,b]$.	**6**
14	Using the Squeeze Theorem, prove that: (a) if $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, then $f \in \mathcal{R}[a,b]$; (b) if $f : [a,b] \to \mathbb{R}$ is monotone on $[a,b]$, then $f \in \mathcal{R}[a,b]$.	**6**
15	State and prove the First Form of the Fundamental Theorem of Calculus.	**6**
16	(a) If $g(x) = x$ for $\lvert x\rvert \geq 1$ and $g(x) = -x$ for $\lvert x\rvert < 1$, and if $G(x) = \frac{1}{2}\lvert x^2 - 1\rvert$, use the Fundamental Theorem of Calculus to show that $\int_{-2}^{3} g(x)\,dx = G(3) - G(-2) = \frac{5}{2}$. (b) Let $B(x) = -\frac{1}{2}x^2$ for $x < 0$ and $B(x) = \frac{1}{2}x^2$ for $x \geq 0$. Show that $\int_a^b \lvert x\rvert\,dx = B(b) - B(a)$.	**6**
17	State and prove the Second Form of the Fundamental Theorem of Calculus: Let $f \in \mathcal{R}[a,b]$ and let f be continuous at a point $c \in [a,b]$. Prove that the indefinite integral $F(z) = \int_a^z f$ for $z \in [a,b]$ is differentiable at c and $F'(c) = f(c)$.	**6**
18	(a) State the Substitution Theorem for Riemann integration and use it to evaluate $\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$. (b) Let F, G be differentiable on $[a,b]$ and let $f = F'$ and $g = G'$ belong to $\mathcal{R}[a,b]$. Prove that $\int_a^b fG = [FG]_a^b - \int_a^b Fg$.	**6**
	Section C	
	Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)	

19	(a) State and prove Cauchy's Criterion for Riemann integrability. (b) Using Cauchy's Criterion, show that Dirichlet's function defined on $[0,1]$ by $f(x)=1$ for $x \in [0,1]$ rational and $f(x)=0$ for $x \in [0,1]$ irrational is not Riemann integrable on $[0,1]$.	**10**
20	(a) Let $f:[a,b] \to \mathbb{R}$ and let $c \in (a,b)$. Prove that $f \in \mathcal{R}[a,b]$ if and only if its restrictions to $[a,c]$ and $[c,b]$ are both Riemann integrable, and in this case $\int_a^b f = \int_a^c f + \int_c^b f$. (b) If $f \in \mathcal{R}[a,b]$ and $[c,d] \subseteq [a,b]$, prove that the restriction of f to $[c,d]$ is in $\mathcal{R}[c,d]$.	**10**